ONE-PAGE BRIEF

# Post-quantum cryptography review: trustwallet-wallet-core @ adf80f8bac98

Client: Trust Wallet — public repository (passive review) · Source: github.com/trustwallet/wallet-core, commit `adf80f8bac98` · Reviewed: 2026-10-29 05:59 UTC · Risk profile: Crypto exchange, custodian or wallet

POST-QUANTUM MATURITY
**Level 0 of 4**
no post-quantum cryptography

FINDINGS
**0** Critical **0** High **9** Medium
**0** Low **15** Info

QUANTUM TIMELINE (MOSCA: X + Y − Z)
Key establishment: 10 + 5 − 8 = 7 → **too late**
Signatures: 10 + 5 − 8 = 7 → **too late**

**Next step:** Inventory where classical public-key cryptography protects long-lived data and keys, and choose the first migration: usually a hybrid TLS key exchange and wrapped keys.

## WHAT MATTERS MOST

| | |
|---|---|
| Medium<br>QSI-2026-06464 | **Transaction signing for 43 chain(s) relies on elliptic-curve signatures** confirmed by the analyst<br>What to do: Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains... |
| Medium<br>QSI-2026-06456 | **Quantum-vulnerable ECDSA: transaction and message signing (84 file(s))** confirmed by the analyst<br>What to do: Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains... |
| Medium<br>QSI-2026-06457 | **Quantum-vulnerable Ed25519: transaction and message signing (33 file(s))** confirmed by the analyst<br>What to do: Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains... |
| Medium<br>QSI-2026-06458 | **Quantum-vulnerable ECDSA: role not determined (18 file(s))** confirmed by the analyst<br>What to do: Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form... |

5 more in the full report.

## WHAT THE QUANTUM-VULNERABLE CRYPTOGRAPHY PROTECTS

| Role | Algorithms | What to do |
|---|---|---|
| Medium transaction and message signing | ECDSA, Ed25519, BLS12-381, Schnorr (BIP-340) ... | Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum... |
| Medium authentication and identity (certificates, tokens, device attestation) | ECDSA | Keep credentials short-lived and plan ML-DSA certificates, tokens and keys as the ecosystem supports them |
| Medium role not determined | ECDSA, Ed25519, X25519, ECDH | Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205)... |

## KNOWN VULNERABLE COMPONENTS

Advisories for the versions in use: 7; unlikely to apply to this build: 7 (a Cargo feature that is off, affected functions not named in the code, or a platform that is not built); to review: 0.

Analyst review: 11 confirmed, 13 not applicable, 0 dismissed as false positives. (QSI review, 2026-10-28)

Static review of source code and manifests; files analysed: 1695. No requests were sent to the owner's systems.

Full report, content SHA-256: `4704b49e843bb279b44c96a91beee1357d25d6854946a2e9dd09694609f0f746`

Full report signature (report.sig.json): Ed25519 key, SHA-256 fingerprint `5b3cf9c9bd0fa54447e2b5dfa7a6bab569bd0c666fa46949599e5c348d17083e`. bundle.sig.json lists the SHA-256 of every file in this bundle and is signed too; verify-bundle checks it offline.

Preliminary findings need confirmation in the code; the full report gives the evidence, the limits of each check and how to re-test.